

Paper Title

Author Name

Abstract—

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I. INTRODUCTION

Deep learning has achieved strong performance in many radar applications, including target detection, classification, and recognition. Its performance, however, depends strongly on the quantity, diversity, and annotation quality of the available data. This dependence is particularly restrictive for maritime radar because existing measured datasets cover only a limited set of sea states, radar parameters, viewing geometries, and target conditions. A model trained on such data may not generalize to an unobserved wind speed, wave direction, grazing angle, or clutter regime. Expanding a measured dataset is also difficult: sea experiments require suitable platforms, radar equipment, environmental sensors, synchronized observations, and favorable weather, while a desired sea condition cannot be reproduced on demand. The shortage and limited diversity of measured radar data have therefore become a major constraint on the development of data-driven maritime sensing methods.

Numerical simulation provides a practical way to supplement measured data because environmental and radar parameters can be controlled independently and a large number of labeled samples can be generated repeatedly. For radar applications, however, computational efficiency alone is insufficient. A simulated sea surface must preserve the statistical roughness of the background sea, respond reasonably to different wind and wave conditions, and represent the local structures that produce strong or non-Gaussian radar returns. High-fidelity hydrodynamic methods can resolve nonlinear wave evolution and breaking interfaces in detail, while full-wave electromagnetic methods can describe the associated diffraction and multipath interactions. Their volume-mesh evolution, fine spatial discretization, or global electromagnetic interactions nevertheless impose substantial computational and storage costs, making repeated simulation over many environmental and radar conditions difficult. Spectral sea-surface and facet-based scattering models are considerably more efficient, but a conventional linear realization does not explicitly represent strongly nonlinear or overturning crests. A useful data-generation method must therefore balance physical credibility, environmental adaptability, computational cost, and compatibility with radar scattering calculations.

Sea-surface roughness is inherently multiscale. Long gravity waves determine the large-scale tilt and motion of the surface, whereas shorter wind waves modify local slopes and radar-resonant components. Consequently, the observed return depends jointly on sea state, incidence angle, radar wavelength, and surface geometry. High-resolution measurements show that sea clutter may exhibit pronounced non-Gaussian tails

and intermittent sea spikes, which can be confused with target responses [?], [?], [?], [?], [?]. Directional wave spectra, such as the unified Elfouhaily spectrum, provide an efficient description of the gravity and gravity-capillary energy distribution [?]. Nevertheless, a linear spectral realization is approximately Gaussian and mainly preserves second-order statistics. Nonlinear hydrodynamic interactions sharpen crests, smooth troughs, and change the distributions of slopes and curvatures, thereby influencing the scattering coefficient and Doppler spectrum [?]. The sea-surface skewness function and its small-scale-wave parameter also affect the upwind-downwind asymmetry of radar backscattering [?], while convolutions of short-wave spectral components can contribute appreciably to the normalized backscattering cross section under different radar frequencies, wind speeds, and incidence angles [?]. Thus, a radar-oriented sea model should represent not only the total spectral energy but also nonlinear crest geometry and the directional organization of shorter wind waves.

The return from an ordinary rough sea surface is commonly described by quasi-specular, Bragg, or two-scale scattering formulations. Their relative contributions vary with incidence angle, and both quasi-specular and Bragg mechanisms may be required in the transition range [?]. Radar observations of marine film slicks further suggest that the source of short wind waves can depend on propagation direction and cannot always be represented by conventional wind-input terms alone [?]. For steep and breaking waves, additional scattering mechanisms become important. Numerical and experimental studies have related low-grazing-angle sea spikes to steepening or breaking crests [?], [?], [?]. Full-wave and hybrid models have investigated crest diffraction, rough-face scattering, and multipath interactions [?], [?], [?], [?], [?]. Facet-based methods have introduced breaking-wave contributions into larger sea surfaces and analyzed their effects on normalized radar cross section (NRCS), sea spikes, and synthetic aperture radar images [?]. Three-dimensional plunging-breaker models combined with ray tracing have also shown that a breaker crest can generate target-like peaks in radar cross section and high-resolution range profiles [?].

Existing approaches present two complementary limitations. On the electromagnetic side, rigorous or high-order numerical methods can resolve diffraction, rough-face scattering, and multipath interactions from breaking waves, but they require fine discretization and repeated solution of large scattering systems [?], [?]. At X-band, conventional numerical discretization may require surface samples much smaller than the radar wavelength, making time-evolving and large-area calculations particularly expensive [?]; even accelerated surface-scattering methods report substantial costs for multiview or full-polarimetric calculations [?]. Lower-cost electromagnetic

approximations also have restricted validity. The traditional two-scale model depends on a prescribed separation between quasi-specular and Bragg regimes and loses accuracy where the two mechanisms coexist [?], whereas steep or multivalued breaker faces introduce crest diffraction, shadowing, and multipath contributions that are not represented by an ordinary rough-surface term alone [?], [?], [?]. Thus, electromagnetic fidelity and computational efficiency remain difficult to obtain simultaneously for large-scale radar-data generation.

On the sea-surface side, computationally efficient spectral and replacement-based methods have different limitations. Linear spectral synthesis preserves the prescribed second-order spectrum but cannot reproduce crest–trough asymmetry or overturning geometry, while Creamer- or Lie-type nonlinear surfaces improve slopes and higher-order statistics but generally remain single-valued height fields [?], [?]. Existing breaking-wave studies often use a selected LONGTANK stage, a measured breaker profile, or a prescribed analytical three-dimensional breaker and then insert or superimpose that geometry into the scattering scene [?], [?], [?], [?]. These constructions are effective for investigating the radar signature of a particular breaker, but their emphasis is not the automatic generation of diverse surfaces conditioned jointly on wind speed, wind direction, short-wave organization, and local crest state [?], [?]. This limitation is important because radar observations indicate that short-wave generation and damping can depend strongly on propagation direction and surface condition [?]. Directly superimposing an independently generated short-wave field without band separation or renormalization can duplicate spectral contributions already contained in the background surface [?], [?], while replacing a region with a prescribed breaker profile may limit geometric continuity and adaptability across sea states [?], [?], [?]. A rapid dataset-oriented model should therefore connect a statistically constrained background, directionally organized short waves, and locally triggered curling geometry within the same environmental parameterization. Its advantage should be evaluated jointly in terms of physical representativeness, environmental flexibility, radar applicability, and measured computational scalability.

This letter proposes a hierarchical sea-surface simulation method intended for rapid radar-data generation under variable environmental conditions. First, an Elfouhaily directional spectrum and a wind-dependent nonlinear Lie transformation are used to construct a statistically constrained nonlinear background surface. Second, a selected short-wave band is replaced by an equal-energy directional component, forming shorter wind-wave ridges with a prescribed dominant propagation direction without repeatedly injecting spectral energy. Third, candidate crests are identified from elevation, propagation-direction slope, and curvature, after which a compactly supported coordinate transformation generates a localized forward-extending and downward-curling breaker. Wind speed, wind direction, directional spreading, short-wave scale, and local curling parameters can be varied to produce different sea-surface realizations, while the facet representation allows the generated surfaces to be used directly in radar scattering and echo simulations. Because the nonlinear and

curling operations are applied through spectral and sparse local surface updates, the method avoids repeated three-dimensional fluid evolution and dense full-wave interaction for every generated realization. It is therefore designed to retain radar-relevant breaker geometry while remaining suitable for large, controllable synthetic radar datasets.

The method is evaluated from surface geometry to radar observables and computational scalability. The background and short-wave-enriched surfaces are first examined using spectral consistency and along-wind and crosswind mean-square slopes under different wind conditions. The localized breaker is then assessed using crest geometry, facet-orientation statistics, radar-facing projected area, and paired unit-footprint scattering enhancement. Radar echoes generated from the complete surface are evaluated through range–time intensity, temporal correlation, and amplitude-distribution statistics. Finally, the measured runtime is decomposed into surface generation, facet-level scattering, echo formation, and input/output costs; its scaling with surface resolution and dataset size is compared with reported X-band numerical methods, with differences in output scope stated explicitly [?], [?], [?]. Section II presents the hierarchical surface model. Section III describes the validation metrics and radar echo generation. Section IV reports the surface and radar comparisons. Section V analyzes computational cost and dataset scalability, and Section VI concludes this letter.

II. HIERARCHICAL SEA-SURFACE MODELING

The proposed framework maps environmental conditions and random realizations to radar-ready dynamic sea surfaces. The environmental input is $\mathbf{p}_e = \{U_{10}, \phi_w, \Omega_c, H_s, L_x, L_y, \Delta x, \Delta y\}$, where U_{10} , ϕ_w , Ω_c , and H_s specify the sea state, while the remaining quantities define the computational domain. A random variable set ξ controls the Fourier phases, allowing independent samples to be generated under identical environmental conditions. Radar parameters are introduced only after the surface has been triangulated, so the sea state and observation configuration can be varied independently.

The complete generation chain is summarized as

$$(\mathbf{p}_e, \xi) \xrightarrow{\mathcal{G}_{NL}} \eta_N \xrightarrow{\mathcal{D}_s} \eta_D \xrightarrow{\mathcal{C}} \mathcal{M}_C \xrightarrow{\mathcal{R}(\mathbf{p}_r)} \mathbf{y}, \quad (1)$$

where $\mathcal{G}_{NL}$ denotes nonlinear spectral background generation, $\mathcal{D}_s$ denotes equal-energy directional short-wave replacement, $\mathcal{C}$ denotes localized curling and triangulation, and $\mathcal{R}$ denotes facet-based radar simulation. The first two operators act over the full domain using Fourier transforms, whereas the curling operator is restricted to selected steep crests. This separation retains large-area statistical control while limiting the additional geometric computation to radar-relevant local structures.

A. Nonlinear Background and Directional Wind Waves

The first modeling stage combines a nonlinear broadband background with an equal-energy directional reorganization of

Placeholder for the overall framework figure

Environmental parameters and random phases → Elfouhaily spectral sampling and nonlinear Lie transformation → equal-energy directional short-wave replacement → steep-crest detection and localized curling → triangular facets and radar echoes.

Fig. 1. Overall framework of the proposed sea-surface and radar-echo simulation method.

its resolved short-wave band. The background supplies sea-state-dependent spectral statistics and non-Gaussian crest geometry, while the reorganized band produces short wind-wave ridges with a controllable dominant propagation direction without duplicating energy already present in the Elfouhaily spectrum.

Let $k = (k_x^2 + k_y^2)^{1/2}$ and $\varphi = \text{atan2}(k_y, k_x)$. The directional elevation spectrum is written as [?]

$$\Psi_\eta(k, \varphi) = \frac{B_l(k) + B_h(k)}{k^4} D(k, \varphi - \varphi_w), \quad (2)$$

where B_l and B_h are the gravity- and capillary-wave curvature spectra and D is the Elfouhaily spreading function. On the discrete wavenumber grid, the linear coefficients are sampled as

$$\hat{\eta}_L(\mathbf{k}) = [\Psi_\eta(\mathbf{k}) \Delta k_x \Delta k_y]^{1/2} \xi(\mathbf{k}), \quad (3)$$

where ξ is a unit-variance complex Gaussian field. Hermitian symmetry is imposed explicitly, and the same random phases are retained during the nonlinear transformation.

To express the Lie correction relative to the wind, the along- and cross-wind wavenumbers are defined as $k_a = k_x \cos \varphi_w + k_y \sin \varphi_w$ and $k_c = -k_x \sin \varphi_w + k_y \cos \varphi_w$. The corresponding Riesz fields are

$$\hat{h}_a = -i \frac{k_a}{k} \hat{\eta}_L \mathcal{B}_i, \quad \hat{h}_c = -i \frac{k_c}{k} \hat{\eta}_L \mathcal{B}_i, \quad (4)$$

where $\mathcal{B}_i$ is the input-band mask. Following the second-order Lie/Creamer construction [?], the nonlinear correction is

$$\begin{aligned} \hat{L} = & -\frac{k_a^2}{2k} \mathcal{F}\{h_a^2\} - \frac{k_a k_c}{k} \mathcal{F}\{h_a h_c\} \\ & - \frac{k_c^2}{2k} \mathcal{F}\{h_c^2\}, \end{aligned} \quad (5)$$

and the preliminary nonlinear spectrum becomes

$$\hat{\eta}_{N,0} = \hat{\eta}_L + \beta \mathcal{P}_H \left\{ \mathcal{B}_o \hat{L} \right\}, \quad (6)$$

where β is dimensionless, $\mathcal{B}_o$ is the output-band mask, and $\mathcal{P}_H$ denotes Hermitian projection. The Riesz multipliers and β are dimensionless; hence, no dimensional wind-speed factor is directly multiplied by the elevation correction. The input and output limits are set to $4k_p$ and $8k_p$, respectively, to retain the resolved quadratic interactions.

Because the nonlinear operation redistributes slope energy, its along- and cross-wind mean-square slopes (MSSs) are

closed against the same Elfouhaily spectrum used for synthesis. For $q \in \{a, c\}$,

$$\begin{aligned} m_q &= \sum_{\mathbf{k}} k_q^2 |\hat{\eta}(\mathbf{k})|^2, \\ G(\varphi) &= \exp \left[\frac{\lambda_0 + \lambda_2 \cos 2(\varphi - \varphi_w)}{2} \right]. \end{aligned} \quad (7)$$

Independent calibration realizations determine smooth wind-dependent along- and cross-wind scale factors by matching the median raw nonlinear MSSs to the integrated Elfouhaily MSSs. For each subsequent realization, λ_0 and λ_2 are solved to realize these fixed scale factors without altering Fourier phase. Cox–Munk and the other empirical slope models are therefore reserved for external validation rather than used as fitting constraints. The closed nonlinear background spectrum is $\hat{\eta}_N = G(\varphi) \hat{\eta}_{N,0}$.

The broadband background already contains short-wave energy, but it does not ensure an organized local propagation direction. A target wavelength interval $[\lambda_{s,\min}, \lambda_{s,\max}]$ is therefore selected, with $k_l = 2\pi/\lambda_{s,\max}$ and $k_h = 2\pi/\lambda_{s,\min}$. Let $W_s(k)$ be a raised-cosine radial window over this interval. The retained background is

$$\hat{\eta}_r(\mathbf{k}) = [1 - W_s(k)] \hat{\eta}_N(\mathbf{k}). \quad (8)$$

For a prescribed propagation unit vector $\mathbf{e}_s = (\cos \psi_s, \sin \psi_s)$, the replacement-band shape is

$$\Phi_s(\mathbf{k}) = W_s(k) \exp \left[-\frac{\log^2(k/k_0)}{2\sigma_k^2} - \frac{\theta_a^2}{2\sigma_\theta^2} \right], \quad (9)$$

where $k_0 = \sqrt{k_l k_h}$, $\sigma_k = \log(k_h/k_l)/4$, and $\theta_a = \cos^{-1}(|\mathbf{k} \cdot \mathbf{e}_s|/k)$. The angular term concentrates the band around the prescribed axis and represents the directional organization observed for short wind waves [?].

A preliminary replacement spectrum is sampled as $\hat{\eta}_{s,0} = \mathcal{P}_H \{\sqrt{\Phi_s} \xi_s\}$. Its scale is determined from the removed-band energy:

$$\alpha_s = \left[\frac{\sum_{\mathbf{k}} |W_s \hat{\eta}_N|^2}{\rho_s \sum_{\mathbf{k}} |\hat{\eta}_{s,0}|^2} \right]^{1/2}. \quad (10)$$

With $\rho_s = 1$, the replacement and removed bands have equal variance by Parseval's theorem. This is a spectral reorganization rather than an additive energy injection.

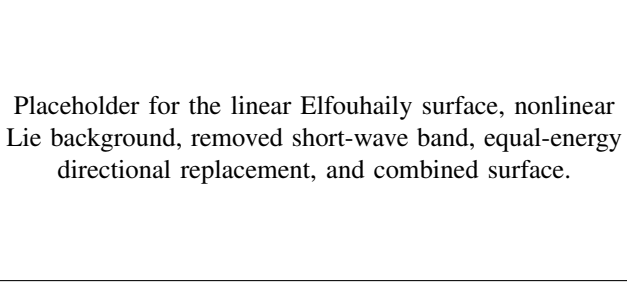


Fig. 2. Construction of the nonlinear background with directionally organized short wind waves.

Unidirectional evolution is introduced by assigning opposite temporal phases to the Hermitian wavenumber pair. The combined surface is

$$\eta_D(\mathbf{x}, t) = \text{Re} \mathcal{F}^{-1} \left\{ \hat{\eta}_r(\mathbf{k}) e^{-i\omega(k)t} + \alpha_s \hat{\eta}_{s,0}(\mathbf{k}) e^{-i\omega(k) \text{sgn}(\mathbf{k} \cdot \mathbf{e}_s) t} \right\}, \quad (11)$$

where $\omega(k) = \sqrt{gk}$ for the resolved short gravity waves. Unless otherwise stated, $\lambda_{s,\min} = 1.5$ m, $\lambda_{s,\max} = 5.0$ m, $\sigma_\theta = 8^\circ$, and $\rho_s = 1$. The minimum wavelength is represented by at least six grid samples. The resulting η_D retains the nonlinear broadband background while providing directionally organized short crests for the local curling stage.

B. Localized Crest Detection and Curling Geometry

The nonlinear directional surface η_D remains single valued and therefore cannot represent a forward-overturning crest. A localized parametric transformation is consequently applied only to selected steep crests. The construction is intended to retain radar-relevant upper-surface geometry at substantially lower cost than resolving the complete air–water flow; it does not model entrained air, spray, or the full hydrodynamic evolution of breaking waves. Similar finite breaker geometries have been used to study localized electromagnetic signatures [?], [?], [?].

Let $\mathbf{e}_s = (\cos \psi_s, \sin \psi_s)$ denote the dominant propagation direction and $\partial_u = \mathbf{e}_s \cdot \nabla$. Candidate crests are restricted to the set

$$\mathcal{D}_c = \{ \mathbf{x} : \eta_D \geq h_q, \partial_u^2 \eta_D < 0, |\partial_u \eta_D| \leq s_c \}, \quad (12)$$

where h_q is a prescribed elevation quantile and s_c is a near-crest slope tolerance. Edge points that cannot contain the complete curling support are excluded. Within $\mathcal{D}_c$, the event center is selected using

$$\mathbf{x}_c = \arg \max_{\mathbf{x} \in \mathcal{D}_c} Q(\mathbf{x}), \quad Q = w_h \tilde{\eta} + w_\kappa (-\widetilde{\partial_u^2 \eta_D}) + w_s \tilde{\eta}_s, \quad (13)$$

where tildes denote normalized positive quantities and η_s is the directional short-wave component. This rule favors high crests with negative directional curvature and a locally strong organized short-wave contribution. It is used to place a parameterized breaking event and is not presented as a universal hydrodynamic breaking-onset criterion.

At $\mathbf{x}_c = (x_c, y_c)$, propagation and crestwise coordinates are defined as

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos \psi_s & \sin \psi_s \\ -\sin \psi_s & \cos \psi_s \end{bmatrix} \begin{bmatrix} x - x_c \\ y - y_c \end{bmatrix}. \quad (14)$$

Let $a = L_c/2$ and $b = W_c/2$ be the crestwise half-length and propagation-direction half-width. A compact cosine window is defined by

$$C(q) = \begin{cases} \frac{1}{2}[1 + \cos(\pi q)], & |q| < 1, \\ 0, & |q| \geq 1. \end{cases} \quad (15)$$

To avoid a perfectly regular footprint, a small centerline displacement $u_c(v)$ and width factor $r(v)$ are introduced and $u_r = u - u_c(v)$ is used below. The core and transition supports are

$$\begin{aligned} W_c &= C\left(\frac{u_r}{br(v)}\right) C\left(\frac{v}{a}\right), \\ W_t &= C\left(\frac{u_r}{(b + \ell_t)r(v)}\right) C\left(\frac{v}{a}\right), \end{aligned} \quad (16)$$

where $\ell_t > 0$ extends the transition region beyond the rotation core. Both windows vanish continuously at their boundaries, while W_t permits the neighboring non-overtaken surface to evolve smoothly before returning to the unchanged background.

The approaching crest is first modified without rotation. An asymmetric width

$$\sigma(u_r) = \begin{cases} \sigma_r, & u_r < 0, \\ \sigma_f, & u_r \geq 0, \end{cases} \quad \sigma_f < \sigma_r, \quad (17)$$

produces a broader rear face and a steeper forward face. With σ_e denoting the transition scale, the pre-rotation coordinates are

$$\begin{aligned} z_p &= \eta_D + A_c \exp\left[-\frac{u_r^2}{2\sigma_e^2(u_r)}\right] W_t + A_e \exp\left[-\frac{u_r^2}{2\sigma_e^2}\right] W_t, \\ u_p &= u + A_l \exp\left[-\frac{u_r^2}{2\sigma_e^2}\right] W_t. \end{aligned} \quad (18)$$

Here, A_c controls the local crest correction, whereas A_e and A_l produce the low-amplitude elevation and forward-lean changes across the wider evolution band. This separation prevents the curl from appearing as an isolated bump placed directly at the highest point.

The rotation center is shifted by $d_c < 0$ toward the approaching shoulder, and the local angle field is

$$\theta(u, v) = \theta_{\max} \exp\left[-\frac{(u_r - d_c)^2}{2(0.62b)^2}\right] C\left(\frac{v}{a}\right). \quad (19)$$

The rotation axis is parallel to the crest direction, with pivot coordinates $u_0 = u_c(v) + d_c + d_f$ and $z_0 = \eta_D - d_p$, where d_f is the forward offset and d_p is the pivot depth. Defining $\Delta u = u_p - u_0$ and $\Delta z = z_p - z_0$, the rotated coordinates are

$$\begin{bmatrix} u_R \\ z_R \end{bmatrix} = \begin{bmatrix} u_0 \\ z_0 \end{bmatrix} + \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta z \end{bmatrix}. \quad (20)$$

Placeholder for crest candidates and selected center,
core and transition supports, propagation-direction
profile before and after curling, and the final
triangulated overturning patch.

Fig. 3. Localized curling-wave construction on a selected directional crest. The rotation begins on the approaching shoulder and is blended into the background through a wider transition region.

The final local coordinates are blended with the wider transition support:

$$\begin{aligned} u' &= u_p + W_t(u_R - u_p), & z' &= z_p + W_t(z_R - z_p), \\ v' &= v. \end{aligned} \quad (21)$$

They are mapped back to the global frame by $x' = x_c + u' \cos \psi_s - v' \sin \psi_s$ and $y' = y_c + u' \sin \psi_s + v' \cos \psi_s$. Because the output is the parametric surface $\mathbf{r}_C(u, v) = [x', y', z']^T$ rather than a height function $z = \eta(x, y)$, the geometry can represent locally overlapping horizontal projections.

The core event mask and the actual overturning mask are distinguished as

$$\begin{aligned} M_c &= \mathbf{1}(W_c \geq \tau_c), & M_o &= M_c \mathbf{1}(J_u < 0), \\ J_u &= \mathbf{e}_s \cdot \nabla u'. \end{aligned} \quad (22)$$

M_c identifies facets materially affected by the curling core, whereas $J_u < 0$ identifies a reversal of the propagation-coordinate mapping and therefore provides a direct geometric indication of overturning. The transition support W_t is not counted as the breaking core.

Finally, the original structured-grid connectivity is retained to triangulate both the uncurled and curled surfaces. For each face $f = (i, j, k)$, its area and unit normal are

$$\begin{aligned} \mathbf{e}_{ij} &= \mathbf{r}_j - \mathbf{r}_i, & \mathbf{e}_{ik} &= \mathbf{r}_k - \mathbf{r}_i, \\ A_f &= \frac{1}{2} \|\mathbf{e}_{ij} \times \mathbf{e}_{ik}\|, & \mathbf{n}_f &= \frac{\mathbf{e}_{ij} \times \mathbf{e}_{ik}}{2A_f}. \end{aligned} \quad (23)$$

Preserving the connectivity is essential because retriangulation in the (x', y') projection can incorrectly connect the upper and lower portions of an overturned crest. The paired baseline and curled meshes therefore have identical face indices, allowing changes in local area, orientation, visibility, and radar scattering to be attributed to the curling transformation.

REFERENCES